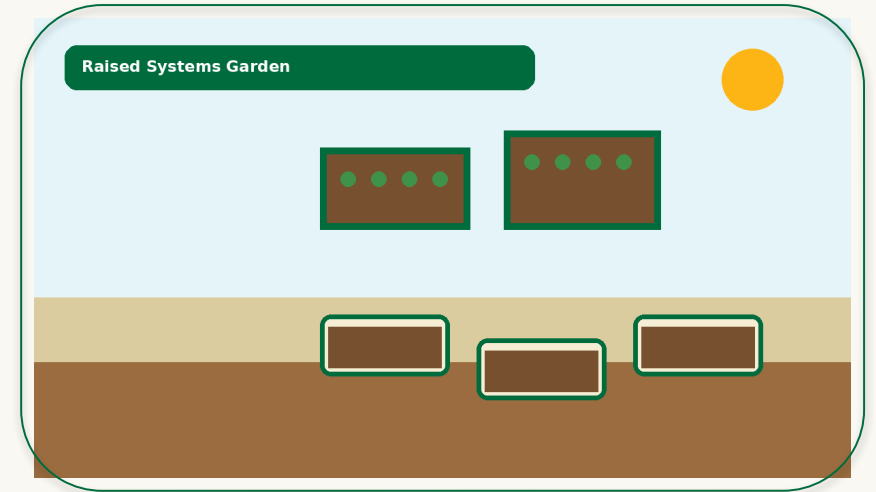


Raised Systems Garden

Comparing Growing Systems

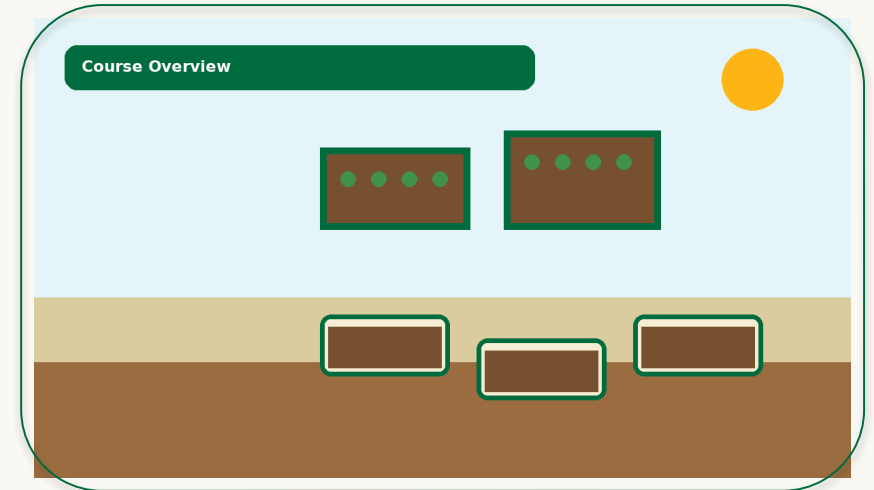
- LIFEWS AgriShine™ GrowMeal™ JSS 1-3 Teacher Training
- Audience: JSS 1-3 teachers, agricultural-science educators and garden facilitators
- Format: Self-paced online course + supervised practical/laboratory orientation
- Recommended duration: 3.5 hours online + 2 hours practical
- JSS learning progression: formulate -> test -> measure -> analyze -> manage



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the production or experimental system; teach all ten technical topics accurately; guide students in planning, measurement and data collection; connect agricultural science with mathematics and evidence; supervise field activities safely; maintain credible records; help students distinguish observation, analysis and conclusion.
- Core materials: teacher guide, plot/system map, crop plan, child-safe field tools, irrigation equipment, labels, clipboards, measuring tapes/rulers, scales and sensors where relevant, data sheets, clean harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical session, and submit a lesson, production plan or experimental protocol appropriate to the course.



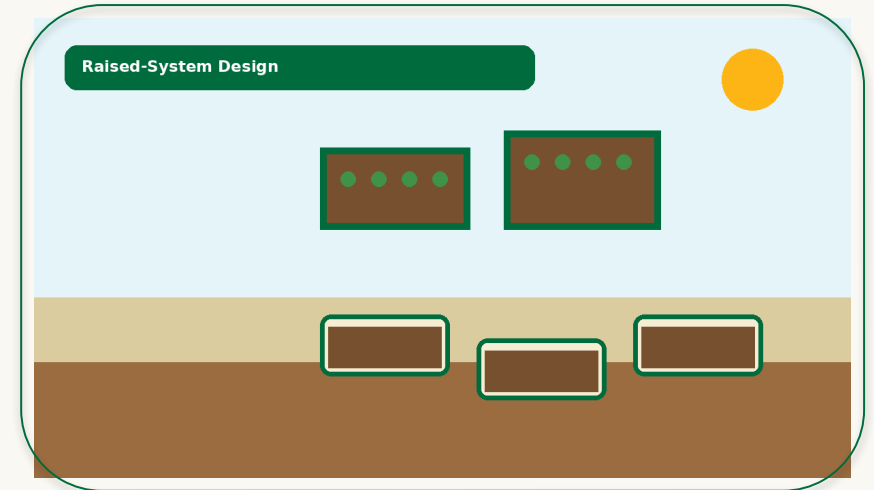
TEACHER PRACTICE

- Training sequence: QUESTION -> PLAN -> IMPLEMENT -> MEASURE -> ANALYZE -> DECIDE.

Raised-System Design

Teacher knowledge, technical practice and student supervision

- Compare raised beds by height, width, materials, accessibility, drainage and intended crop use.
- Relate bed width to safe reach and bed depth to crop rooting requirements.
- Include pathways and irrigation access in design evaluation.
- Assess structural durability and safety before student use.
- Treat design as a set of trade-offs rather than assuming taller or larger beds are better.



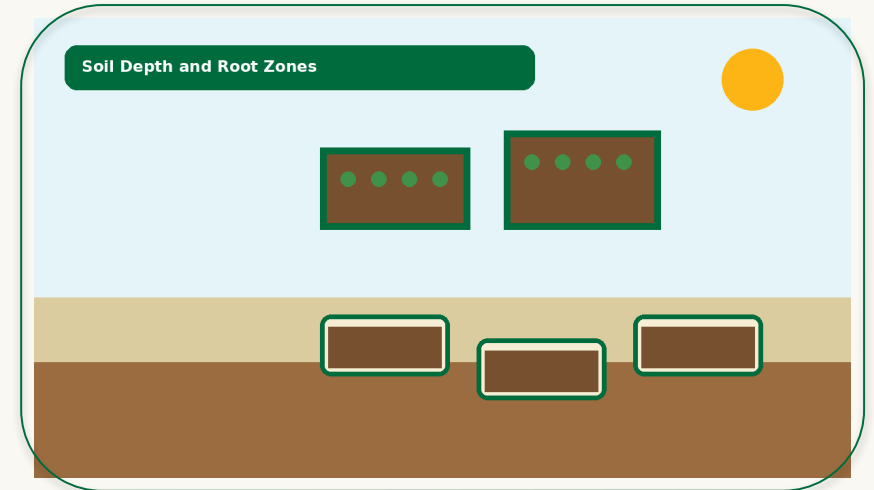
TEACHER PRACTICE

- Evaluate two raised-system designs using a common rubric.

Soil Depth and Root Zones

Teacher knowledge, technical practice and student supervision

- Explain root-zone depth as the volume of growing medium available for roots, water and nutrients.
- Compare shallow- and deeper-rooted crops with the bed depth provided.
- Observe root systems only from planned samples or harvested crops.
- Avoid disturbing productive plants merely to inspect roots.
- Record soil depth consistently across systems.



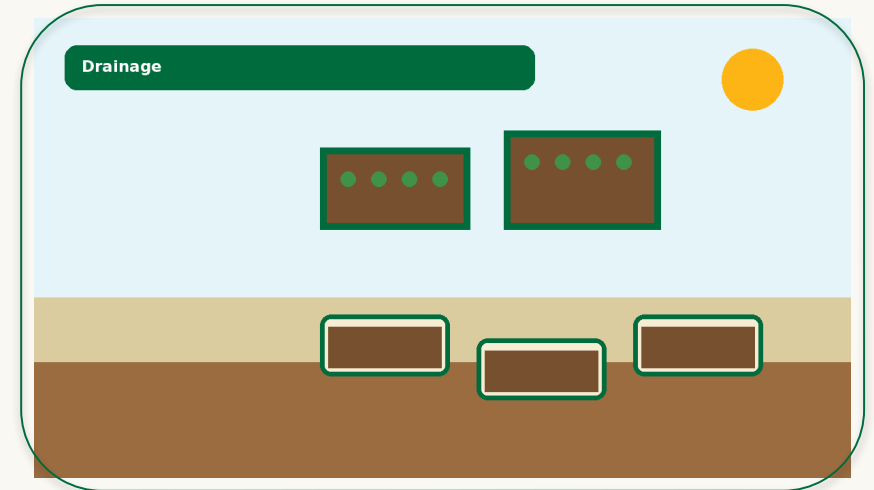
TEACHER PRACTICE

- Measure effective growing depth in two systems.

Drainage

Teacher knowledge, technical practice and student supervision

- Identify drainage outlets, soil/media structure and signs of waterlogging or rapid drying.
- Observe drainage after a controlled irrigation event.
- Explain that good drainage must still retain enough water for crop needs.
- Keep runoff from creating erosion or unsafe pathways.
- Record drainage problems by system.



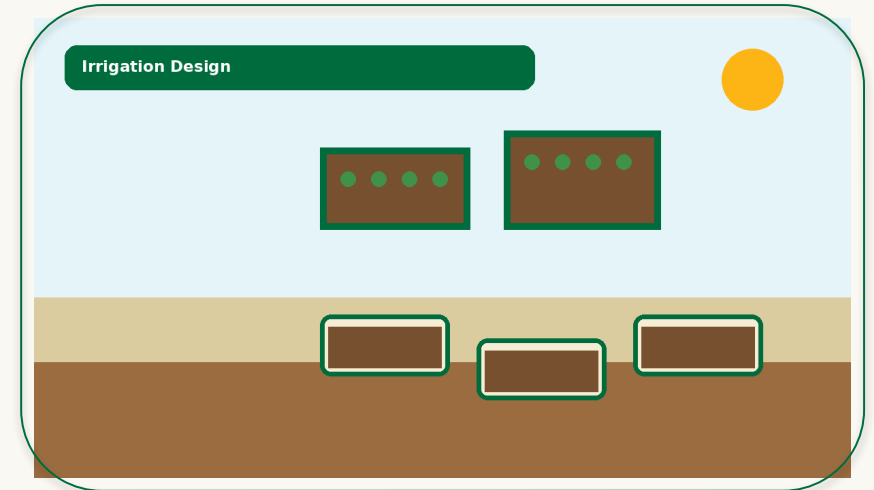
TEACHER PRACTICE

- Prepare a drainage comparison protocol.

Irrigation Design

Teacher knowledge, technical practice and student supervision

- Map how water enters and moves through each raised system.
- Compare hand watering, drip lines or other approved methods using consistent criteria.
- Consider emitter placement, pressure/flow, bed dimensions and crop arrangement.
- Measure water delivery where feasible.
- Record leaks, dry zones and maintenance requirements.



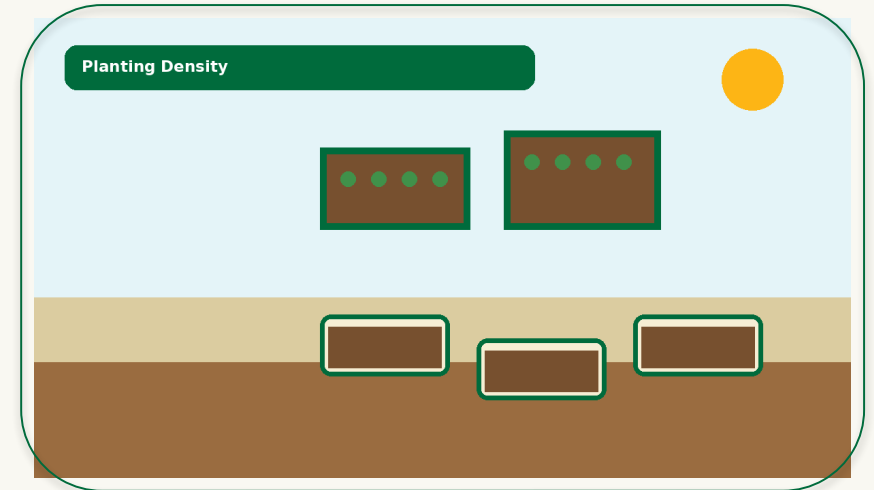
TEACHER PRACTICE

- Draw the irrigation layout for two raised systems.

Planting Density

Teacher knowledge, technical practice and student supervision

- Calculate target and actual plant density for each system.
- Use crop-specific spacing rather than equal plant numbers when bed areas differ.
- Relate density to canopy development, access and competition.
- Record thinning or mortality.
- Use density as one explanatory variable when comparing performance.



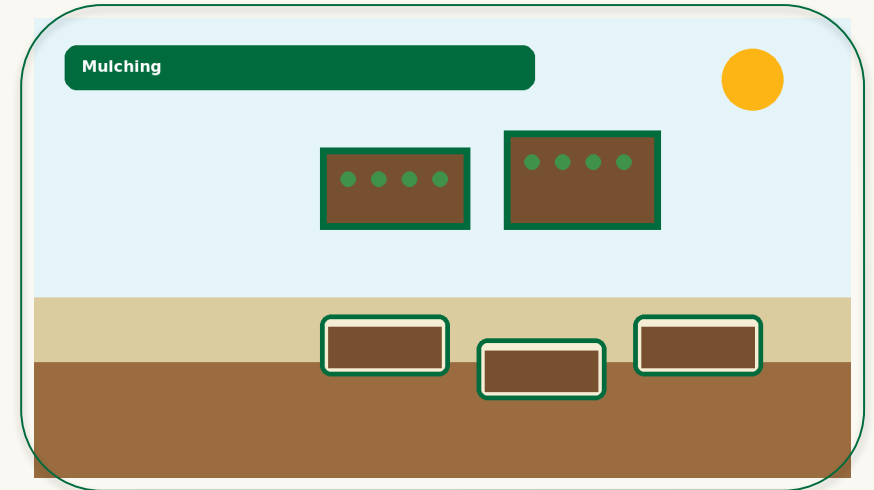
TEACHER PRACTICE

- Calculate plants per square meter for two beds.

Mulching

Teacher knowledge, technical practice and student supervision

- Apply the same approved mulch material and thickness where mulch is part of a fair system comparison.
- Measure or observe soil moisture and weed presence over time.
- Keep mulch clear of vulnerable stems where appropriate.
- Record application date and amount.
- Avoid assuming mulch effects are identical across all bed designs.



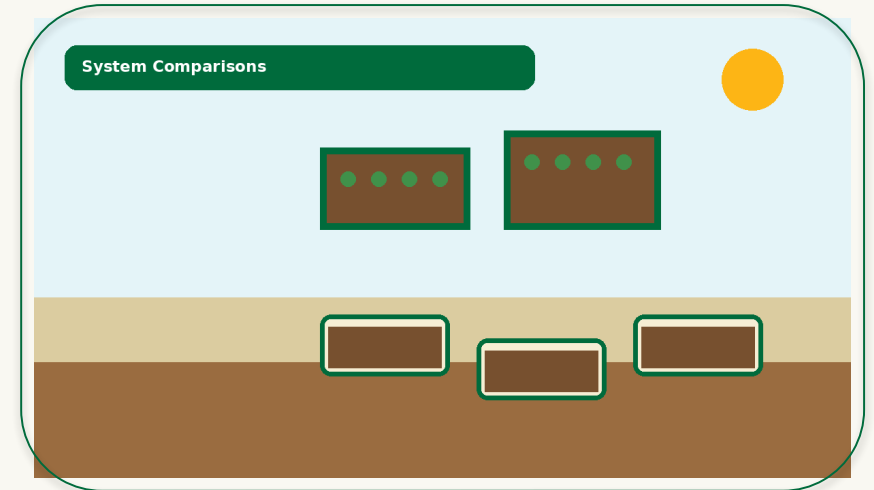
TEACHER PRACTICE

- Set up a controlled mulch observation.

System Comparisons

Teacher knowledge, technical practice and student supervision

- Define the comparison question before collecting data.
- Select common indicators such as water use, soil moisture, growth, labor, yield or maintenance.
- Keep crop, planting date and measurement methods similar where possible.
- Document unavoidable differences between systems.
- Use comparison to identify context-specific strengths and limitations.



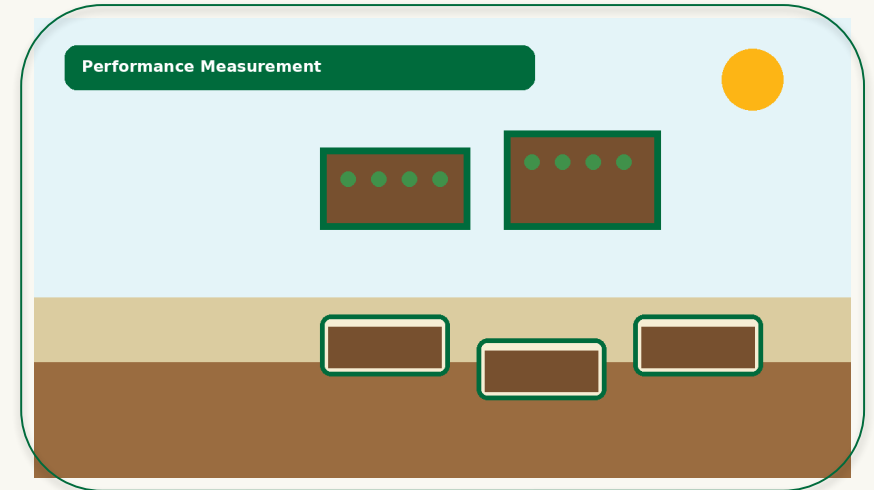
TEACHER PRACTICE

- Write a fair comparison question for two raised systems.

Performance Measurement

Teacher knowledge, technical practice and student supervision

- Measure agreed indicators using consistent tools, units and timing.
- Use multiple sample plants or repeated observations.
- Record raw data before calculating summaries.
- Include resource inputs where relevant, not only crop growth.
- Check unusual measurements before excluding them.



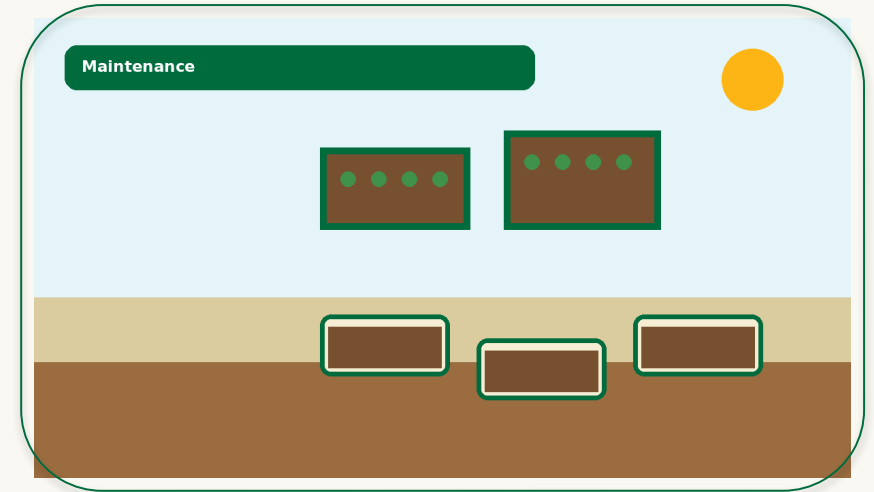
TEACHER PRACTICE

- Prepare a system-performance data sheet.

Maintenance

Teacher knowledge, technical practice and student supervision

- Inspect structural edges, fasteners, soil level, irrigation, drainage, labels and pathways.
- Record maintenance frequency and time where useful.
- Repair unsafe structures before student access.
- Replace damaged irrigation components promptly.
- Include maintenance burden when evaluating a system.



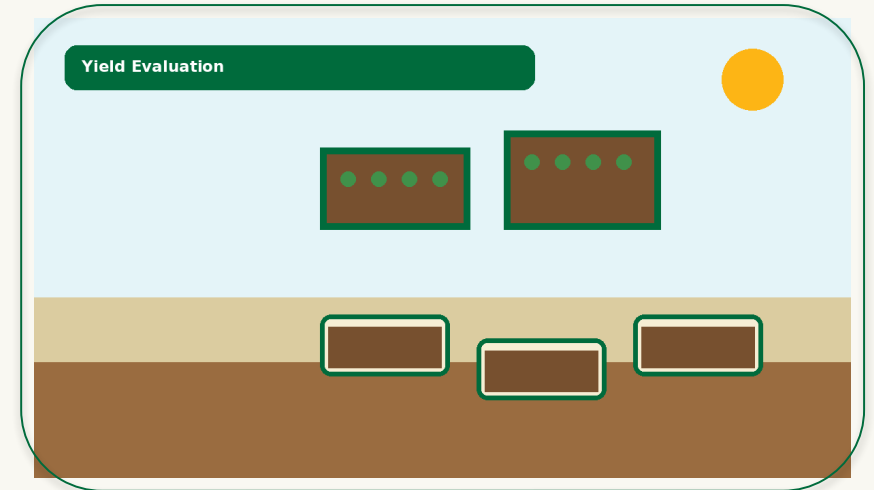
TEACHER PRACTICE

- Create a weekly maintenance log.

Yield Evaluation

Teacher knowledge, technical practice and student supervision

- Measure harvest mass/count from a known bed area.
- Calculate yield per bed and, where appropriate, yield per unit area.
- Compare yield alongside water use, labor and maintenance rather than as the only performance metric.
- Recognize environmental and management differences when interpreting results.
- Use repeated crop cycles before making strong design claims.



TEACHER PRACTICE

- Completion task: prepare a multi-criteria raised-system evaluation.